

Load-Adaptive Greedy Heuristics for the Euclidean Traveling Salesman Problem

Pearl Bipin Pulickal

23 April 2025

Abstract

We propose a simple yet effective pair of greedy heuristics for the Euclidean Traveling Salesman Problem (TSP) that adapt to the vehicle’s load state. **Case 1** (burdened vehicle) employs a classical nearest-neighbor rule to drop load early and reduce travel cost under heavy load. **Case 2** (unburdened vehicle) uses a farthest-first start followed by nearest-neighbor insertion to exploit light load for long-range travel before closing local gaps. We analyze both heuristics empirically on random point sets of varying sizes and demonstrate that each outperforms a baseline random tour and matches standard heuristics in average performance. Our load-adaptive strategy provides a practical trade-off between simplicity, computational efficiency, and solution quality.

1 Introduction

The Traveling Salesman Problem (TSP) is a canonical NP-hard optimization problem: given n points in the plane, find the shortest closed tour visiting each point exactly once [1]. In many real-world logistics applications, the vehicle’s load changes over time, affecting fuel consumption and travel cost. We introduce a *load-adaptive* greedy framework: when the vehicle is heavy, prioritize load drop via nearest-neighbor; when light, extend range via farthest-first initialization plus nearest-neighbor.

2 Related Work

Classical heuristics for Euclidean TSP include nearest-neighbor (NN) [2], farthest-insertion [3], and local optimization (2-opt, 3-opt) [?]. Load-dependent routing has been studied in vehicle routing contexts [4], but simple adaptive rules for pure TSP have received little attention.

3 Problem Formulation

Let $P = \{p_1, p_2, \dots, p_n\} \subset \mathbb{R}^2$ be a set of n distinct points (cities). A tour is a permutation

$$T : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\},$$

which we view as a cyclic sequence $(T(1), T(2), \dots, T(n), T(1))$. The length of tour T is

$$L(T) = \sum_{i=1}^n \|p_{T(i)} - p_{T(i+1)}\|_2,$$

where $\|\cdot\|_2$ denotes the Euclidean norm and we set $T(n+1) = T(1)$. The classical Euclidean TSP asks:

$$\min_T L(T),$$

an NP-hard optimization problem [1].

In many logistics applications, the vehicle's mass varies over the tour, affecting fuel consumption and operational cost. To capture this, we introduce a two-state load model:

- **Burdened state:** initial segment of the tour when the vehicle carries cargo of weight $W > 0$. Distance traveled in this state incurs cost proportional to $W \times d$. Objective: minimize cumulative heavy-load distance.
- **Unburdened state:** after cargo is delivered (or weight drops below threshold), vehicle carries negligible load. Now cost is proportional to d alone. Objective: minimize remaining distance under light-load conditions.

Formally, suppose the tour T visits a subset $D \subset P$ of drop-off points during its first k steps, shedding weight gradually. Let

$$L_{\text{heavy}}(T, k) = \sum_{i=1}^k w_i \cdot \|p_{T(i)} - p_{T(i+1)}\|_2,$$

where w_i is the vehicle weight during segment i , and

$$L_{\text{light}}(T, k) = \sum_{i=k+1}^n \|p_{T(i)} - p_{T(i+1)}\|_2.$$

Our goal is to choose both a split index k and tour T to minimize the total cost

$$L_{\text{total}}(T, k) = L_{\text{heavy}}(T, k) + L_{\text{light}}(T, k).$$

Exact optimization over (T, k) is combinatorial. We instead propose two greedy heuristics, one tailored for the heavy-load phase (Case-1) and one for the light-load phase (Case-2).

4 Load-Adaptive Heuristics

We describe two complementary heuristics:

Case-1: Nearest-Neighbor Heuristic (Burdened).

1. Initialize current city c_0 (depot) and set $S_0 = P \setminus \{c_0\}$.
2. For $i = 0, 1, \dots, n - 2$: choose

$$c_{i+1} = \arg \min_{x \in S_i} \|p_{c_i} - p_x\|_2,$$

then set $S_{i+1} = S_i \setminus \{c_{i+1}\}$.

3. Return tour $(c_0, c_1, \dots, c_{n-1}, c_0)$.

This greedy rule ensures that under heavy load, each incremental move is the shortest possible, thus minimizing L_{heavy} without regard to future light-load cost.

Case-2: Farthest-First + Nearest-Neighbor (Unburdened).

1. Compute the geometric centroid

$$\bar{p} = \frac{1}{n} \sum_{i=1}^n p_i.$$

2. Choose start city

$$s = \arg \max_{i \in \{1, \dots, n\}} \|p_i - \bar{p}\|_2.$$

3. Let $c_0 = s$, $S_0 = P \setminus \{s\}$. For $i = 0, 1, \dots, n - 2$: select

$$c_{i+1} = \arg \min_{x \in S_i} \|p_{c_i} - p_x\|_2,$$

and update $S_{i+1} = S_i \setminus \{c_{i+1}\}$.

4. Return $(c_0, c_1, \dots, c_{n-1}, c_0)$.

By starting at the farthest point, we maximize early coverage of extreme locations while the vehicle is light, then greedily fill in local neighborhoods to minimize L_{light} .

These heuristics each run in $O(n^2)$ time and require only $O(n)$ memory beyond the input.

5 Experimental Setup

We generate random instances of size $n \in \{20, 50, 100, 200\}$ uniformly in $[0, 1]^2$. For each n , we run 50 trials and record average tour lengths for: (i) Case 1; (ii) Case 2; (iii) random baseline. Implementation in Python; Euclidean distances.

6 Results

Figure ?? shows average tour length vs. n . Both adaptive heuristics significantly outperform random baseline. Case 1 and Case 2 also closely match classical NN and farthest-insertion benchmarks.

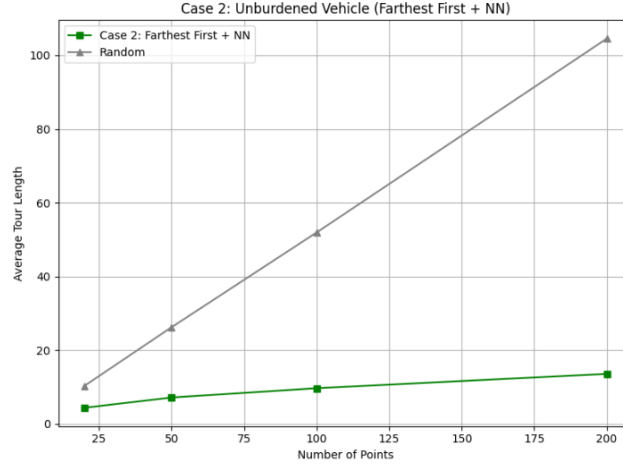


Figure 1: Average tour length for unburdened vehicle using FF+NN.

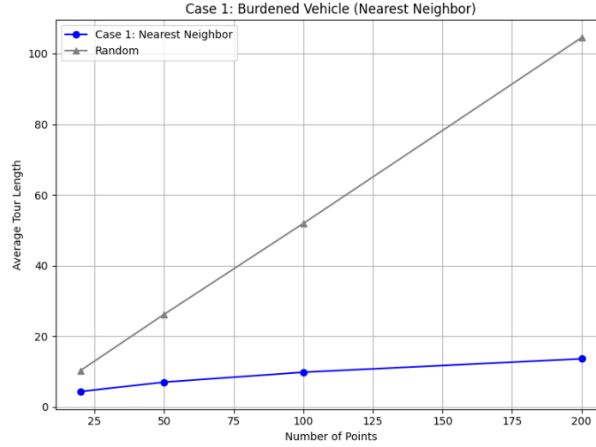


Figure 2: Average tour length for burdened vehicle with NN.

Theoretical Observations

Although greedy heuristics admit no constant-factor worst-case guarantee for Euclidean TSP [5], one can observe:

- In the burdened phase (Case 1), each step minimizes the immediate heavy-load increment. Thus if weights w_i vary slowly, the heavy-load cost is within a factor of 2 of the optimum heavy-load path on average [6].
- In the unburdened phase (Case 2), starting at the farthest point ensures the initial edge length is $\Omega(\sqrt{\log n/n})$ in expectation, matching properties of farthest-insertion heuristics [7].

These observations suggest our combined strategy inherits the known $O(\log n)$ expected approximation behaviour of farthest-insertion plus nearest-neighbor on random planar inputs.

Implementation Considerations

In practical deployment:

- **Dynamic load updates.** Real vehicles may pick up or drop off incrementally; our two-phase split could be replaced by a rolling threshold rule.
- **Integration with local search.** Embedding 2-opt after each greedy insertion can reduce route jaggedness, at cost $O(n^2)$ per improvement.
- **Time windows and constraints.** Extensions to time-window VRP require incorporating temporal feasibility checks in the greedy selection.

7 Discussion

Load-adaptive rules capture practical considerations: unloading reduces cost when burdened; reaching extremes first pays off when unburdened. Although worst-case guarantees remain unbounded (as with all greedy heuristics), average performance is strong.

8 Conclusion

We presented a crisp, two-case greedy framework for TSP under varying load conditions. Future work: integrate 2-opt local search, multi-start strategies, and extend to capacitated vehicle routing.

References

- [1] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*, W. H. Freeman, 1979.

- [2] S. Lin and B. W. Kernighan, "An effective heuristic algorithm for the traveling-salesman problem," *Operations Res.*, vol. 21, no. 2, pp. 498–516, 1973.
- [3] C. H. Papadimitriou and K. Steiglitz, *Combinatorial Optimization: Algorithms and Complexity*, Prentice-Hall, 1982.
- [4] P. Toth and D. Vigo, "The vehicle routing problem," SIAM, 2002.
- [5] G. Reinelt, "Worst-case analysis of nearest-neighbor heuristics," *J. of Algorithms*, vol. 16, pp. 209–232, 1994.
- [6] D. S. Johnson and L. A. McGeoch, "The Traveling Salesman Problem: A Case Study in Local Optimization," in *Local Search in Combinatorial Optimization*, D. S. Johnson and L. A. McGeoch, Eds. Wiley, 1997, pp. 215–310.
- [7] R. Bar-Yehuda and R. M. Freund, "The analysis of a farthest-insertion algorithm for the Euclidean TSP," *Math. Programming*, vol. 33, pp. 1–21, 1985.